

Gulkaran Singh

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EXPERIENCE

Shopify

Sept 2025 - Present

Software Engineer

Toronto, ON

- Drove significant enhancements across Shopify's new core release-management platform by shipping new features, optimizing command performance by over **20%**, and refactoring critical components to accelerate release engineering workflows impacting the deployments of **thousands** of engineers.
- Led the end-to-end development of a critical **deployment notifications system**, from architecting the initial proof-of-concept to launching the production service that delivered **8,000+ Slack alerts** to nearly 2,500 engineers in its first two weeks at sub-second latency.

Software Engineering Intern

May 2025 - Sept 2025

- Drove significant enhancements across Shopify's new core release-management platform by shipping new features, optimizing command performance by over **20%**, and refactoring critical components to accelerate release engineering workflows impacting the deployments of **thousands** of engineers.
- Led the end-to-end development of a critical **deployment notifications system**, from architecting the initial proof-of-concept to launching the production service that delivered **8,000+ Slack alerts** to nearly 2,500 engineers in its first two weeks at sub-second latency.

Scotiabank

Jan 2024 - Apr 2024

Software Developer Intern

Toronto, ON

- Designed frontend components and backend systems for **3+** projects leveraging **C#, Vue.js, SQL, Javascript, and jQuery** for capital market teams through **agile** development.
- Centralized individual trade floor visitation requests into a bulk import/approval dashboard, using **ClosedXML, SQL, and Vue.js**, increasing processing speed by **70%**, enabling faster response times.
- Developed a condensed, editable dashboard that merged new operational resilience workflows into an existing system by leveraging **SQL transactions** and **rollbacks**, reducing data processing time by **50%**.

PROJECTS

Fluid Simulation Engine | Rust, Bevy, Cargo

- Engineered a real-time 2D fluid simulation in **Rust** using the **Bevy** game engine, implementing Smoothed Particle Hydrodynamics (SPH) for 1,500+ particles with density estimation, pressure forces, spatial grid optimizations and elastic collisions.

Handtex - AI Image Detection | Python, React, Typescript, TensorFlow, OpenCV

- Designed a convolutional **neural network** with **TensorFlow**, utilizing **machine learning** to achieve a **97% accuracy** in predicting mathematical symbols from images, integrated into a **React** and **Flask API** app.

Markdown Documentation Site | AWS (S3, DynamoDB, Lambda, API Gateway), React, Typescript, Next 13

- Created a dynamic documentation site with **CI/CD automation** with **Next 13** and **AWS** for seamless file upload to **S3** and synchronization with **DynamoDB** to showcase completed LeetCode problems.

EDUCATION

McMaster University

Sept 2022 - Apr 2026

Bachelor of Applied Science in Computer Science, Minor in Business

GPA: 3.95/4.0

TECHNICAL SKILLS

Languages: Python, Ruby, JavaScript, C#, Java, Typescript, SQL, C, Rust, C++, HTML, CSS, LaTeX, MATLAB, R

Libraries & Frameworks: React.js, Vue.js, Next.js, Ruby on Rails, Node.js, Three.js, jQuery, Bootstrap, Tailwind, CSS, ASP.NET, Spring Boot, Flask, Tensorflow, Socket.IO, Bevy

Data: PostgreSQL, Microsoft SQL Server, Firebase (Firestore), MongoDB

Cloud: AWS, S3, DynamoDB, Lambda, API Gateway, ELB, Kafka

Other: Docker, Kubernetes, GitHub Actions, Linux, Git, Postman, MS Office (Excel, Teams, Word, PowerBI)